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A Case Study: Analysing the Relationship Between Regional Rainfall and Water Storage in Cape Town's 'Big Six' Dams

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Abstract

In response to Cape Town's "Day Zero" water crisis and calls for the integration of seasonal forecast into process the factors influencing the prediction of water storage in the Western Cape are assessed. Where spatial distributional analysis of daily precipitation shows the variable influence of rainfall regimes across the region, highlighting the need for considering seasonal forecasts at an appropriate scale. Further whilst it may seem intuitive that rainfall and streamflow are closely correlated, being that rainfall is the sole input of water into the system, the relationship is shown to be far more complicated whereby the relationship is shown to be highly seasonal, with increased correlations occurring in winter and decreased correlations in summer. Streamflow is seen to lag rainfall with increased streamflow's in autumn compared to spring. Distributional shifts in the rainfall-streamflow relationship are identified such that in autumn a below-normal rainfall forecast better correlates to a not above-normal streamflow, whilst the opposite but to a lesser degree is seen in summer and winter, thus presenting opportunities for increasing the appreciable probabilistic skill of a seasonal forecast.

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Introduction

Drought is an endemic feature of South African history, often resulting in severe and widespread economic and societal consequences (Vogel & Drummond, 1993). Cape Town in specific, is a water scarce region (Ziervogel et al., 2010), a reality that has recently been brought to light with the near catastrophic failure of the City of Cape Town's (CoCT) water supply, where a complete shutdown of municipal water supply was narrowly avoided thanks, in most part, to the success of drastic demand management schemes and cooperation from the public (*Summary – Water Outlook*, 2018).

Seasonal forecasting in the Western Cape is seen to be less skilful than the rest of southern Africa due in large part to the predominantly winter rainfall in the region being reliant on extra-tropical rather than tropical systems (Mason et al., 1996). Further decisions on whether water restrictions are to be implemented are only made once a year, thus only forecasts that provide a 12-month outlook are of use to policy makers, which further reduces the useful skill of these forecasts. In the absence of tailored forecasting products, seasonal precipitation forecasts are considered for use in predicting streamflow into dams by water managers. However, there is little evidence for these forecasts effecting decision making due in part to the low skill of these forecasts (Ziervogel et al., 2010). For this reason, during the 2017-2018 water crisis, rather than using a seasonal forecast, the previous year's rainfall was used as a proxy for use in decision making (*Summary – Water Outlook*, 2018).

The CoCT is heavily reliant on water demand management during dry periods to ensure continued water supply. Where despite a growing city, demand has remained steady since 2000 and severe water restrictions are a common occurrence (Groundup, 2018). Water restrictions reached a peak in 2018 where restrictions of 50L per person per day were implemented. Further demand management is however acknowledged to be unfeasible. demand management places the City in a precarious position should demand management schemes be unsuccessful. Because of this, the City has developed augmentation schemes should water shortages occur again. However, these augmentation schemes are more costly and require notice to be implemented (*Summary – Water Outlook*, 2018).

Wolski et al. (2017) suggest that water resource management could benefit from the development of seasonal forecasts in conjunction with hydrologists. These forecasts would be tailored to providing information specific to water resource management and would have potential to improve the skill of the forecast in comparison to current seasonal precipitation forecasts, thus providing policy makers with a more relatable and understandable tool. Whilst Ziervogel et al. 2010 (pg. 546) identified that water managers in the region expressed that improved seasonal forecast skill,

improved understanding of the mechanisms behind seasonal forecasts and that improved understanding of the skill associated with forecasts are necessary for improving their usefulness to decision makers.

Therefore, the sources of uncertainty in using a regional precipitation forecast for the prediction of water storage in the Western Cape are investigated. With the view to identifying areas where skill could potentially be improved and to serve to contribute to a better understanding of the factors involved in the prediction of water storage in the Western Cape.

Aims and Objectives

Aim

To assess the relationship between rainfall across the Western Cape and water accumulation in Cape Town's big six dams, with the view to identifying key areas where a seasonal precipitation forecast can be improved for the prediction of water supply.

Objectives

- Identify spatial variations in rainfall across the western cape
- Analyse the relationship between rainfall and surface water storage accumulation

Literature Review

Cape Town is Vulnerable to Water Scarcity and Heavily Reliant on Demand Management

South Africa is a water scarce region with semi-arid to arid characteristics (WRC, 2015). Most of the country relies primarily on summer rainfall from convective and tropical systems, which are associated with strong dependencies on ENSO events. However, the Western Cape of South Africa receives predominantly winter rainfall as a result of organized synoptic-scale weather systems originating in the South Atlantic, and as such do not display the same ENSO dependency and have no clear source of predictability (Engelbrecht et al., 2015)(Philippon et al., 2012). Observed temperatures in the Western Cape are increasing (New et al., 2006) and climate projections predict increased rainfall variability and increased temperatures as a result of anthropogenic climate forcing in the South Western Cape region (Hewitson & Crane, 2006). Further, the likelihood of the 2015-2017 Western Cape drought has been demonstrated to have tripled as a result of recent climate change (Otto et al., 2018). Thus, drought management in the Western Cape is of increasing importance.

The 2015-2018 water crisis in Cape Town, where a complete shutdown of municipal water was only narrowly avoided through drastic demand management and augmentation schemes (*Summary – Water Outlook 2018*, 2018), was as a result of a 3 year drought event, an occurrence with an estimated return period of 300 years (Wolski, 2018; Simpkins, 2018). Factors such as increased water demand from a growing City, water resource not keeping pace with growing demand, changing land-use practices and a lack of systemic drought risk planning all served to put further pressure on the Western Cape Water Supply Scheme (WCWSS) (WRC, 15)(Vogel & Drummond, 1993).

As early as 2004, Cape Town was identified by the Department of Water Affairs as the first major metropole where water demand will exceed supply. This was relieved somewhat by the construction of the Berg river dam which was integrated into the WCWSS in 2009 but even with the resulting 25% increase in supply, demand was expected to exceed supply by 2013 if no demand management schemes were implemented. (Mukheibir & Ziervogel, 2007, pg. 150) These demand management schemes have proved very effective with demand remaining relatively steady since 2000 despite the CoCT growing by 38% over the period 2000 to 2016 (Groundup, 2018) (van Zyl, 2017). Furthermore demand has been reduced by almost 25% in the past year as a result of the CoCT's "Day Zero" campaign with the goal to reduce water consumption by a further 12.5%¹ (*Summary – Water*

¹ (Foot note: 800 mL/day December 2016 to 600 mL/day December 2017. 87L per person with goal of 500 mL/day of treated water) (*Summary – Water Outlook 2018*)

Outlook, 2018). It is acknowledged that any further relief through demand management is no longer viable and as such, various water augmentation schemes are being implemented into the WCWSS with the goal to increase potential augmented supply to 500 mL/day by 2022. Potential Augmentation schemes include Spring water integration, desalination, groundwater harvesting and re-use of effluent water (*Summary – Water Outlook*, 2018). The WCWSS is thus seen to be heavily reliant on cooperation from the public to reduce demand, which although so far successful, puts the City in a potentially precarious situation should further demand management no longer be successful.

Low Skill of Seasonal Forecasts in this Region is Widely Acknowledged

Due to the 2015-2018 water crisis reaching a peak in the Autumn of 2018 and the necessity of improved rainfall over the 2018 winter for avoiding a complete failure of water supply becoming clear, seasonal forecasts for the 2018 winter became of increasing importance and under increasing scrutiny. There is little evidence for utilisation of seasonal forecasts amongst decision-makers due to little skill being demonstrated and the difficulties of integrating these forecasts into process (Ziervogel et al., 2010), thus forecasts issued preceding the 2018 winter likely had little impact on policy. These forecasts did however receive widespread attention across mainstream media, it is unclear whether the outcomes of these forecasts had any effect on water demand but due to the reliance from the CoCT on demand management, favourable forecasts could potentially have resulted in a false sense of security in the public domain, and resulted in a subsequent increase in water demand, thus placing the City in a precarious position should the forecast be wrong.

Seasonal forecasts are presented as probabilistic presentations, and thus by definition cannot be wrong as a probability is assigned to each outcome. Generally, these forecasts are presented in tercile form where probability of above-normal, normal or below-normal rainfall events are provided. Here equally proportioned fractions of the percentiles, below the 33rd percentile, below the 67th percentile and above the 67th percentile or rather below-normal, normal or above-normal are considered. Thus rather than providing an absolute indication of expected conditions, seasonal forecasts provide an indication of the expected short term shift in the probability distributions in a regions climate (Johnston et al., 2004). However for a forecast to be usable to a consumer an indication of the accuracy and reliability is needed (Johnston et al., 2004)(Ziervogel et al., 2010). Skill scores provide a means to quantify how often and how close a forecast has in the past been to the actual outcome. For example, the likelihood skill score quantifies the skill through assessing what probability was assigned to the actual outcome and scaling it such that in a tercile system if the actual outcome was forecasted as 0.33 a skill of 0 is assigned and a maximum score of 1 if this outcome was forecast as 100% likely. This method does not however account for what probabilities

where assigned to events that didn't occur. The ranked probability skill score improves on this through measuring the squared forecast probability error of all the forecasted categories and then scaling against a climatological forecast. However, this technique is biased toward neutral events where a normal event has 2 neighbouring outcomes and as such an adjustment is made where if normal is observed the score is multiplied by 2 and if below- or above-normal is observed the score is multiplied by 0.8 (*Descriptions of the IRI Climate Forecast Verification Scores*, 2013). Thus, an important balance of how fully the skill of the model is captured and how easily the skill score can be understood by the user needs to be considered.

There are a number of Seasonal precipitation forecasts available that cover the South Western Cape catchment region. The South African Weather Service's Seasonal Climate Watch is issued on a monthly basis. The latest forecasts produced as of May 2018 regarding the 2018 winter provided contradictory results. The following seasonal forecasts were available; The ensemble EHCAM atmospheric general circulation model which indicated normal conditions; the ensemble SAWS coupled model indicated below normal conditions with 70-100% probability; The SAWS statistically downscaled coupled model contradicted the ensemble SAWS coupled model and suggested above-normal conditions over the whole of South Africa for the upcoming winter rainfall season. No skill scores were available for these forecasts. The expected number of rainfall days greater than both 5mm and 15mm were forecast as above-normal for JJA 2018. The IRI multi-model global forecast indicated a 40-45% probability of above normal conditions occurring in the south western cape. Importantly little to no skill was associated with all of these forecasts with the exception of the >15mm rain days which did display sufficient skill. Whilst it is evident that much confusion and contradiction exist within these forecasts the modal prediction of normal to above normal conditions was much publicised across local news. Whether this had an impact on public water savings is unclear, but the heavy reliance of the CoCT on demand management makes this a potentially dangerous prospect. The winter of 2018 has however proved to be sufficient to end the climatological drought, where the SAWS historical rain map tool² indicates between 75-125% of normal over the main catchment region, and dam levels are now at 75.6% from last year's 38.5%³, nevertheless the consequences of continued drought need be considered.

A lack of skill being a primary reason for the limited uptake of seasonal forecasts into industry in South Africa is acknowledged by Wolski et al., 2017 (Pg.31), Ziervogel et al., 2010 (Pg. 546) and

² Available at <http://www.weathersa.co.za/climate/historical-rain-maps>

³ Available at : <https://resource.capetown.gov.za/documentcentre/Documents/City%20research%20reports%20and%20review/damlevels.pdf>

Johnston et al., 2004 (Pg. 78). Johnston et al., 2004 further acknowledges that “Temporal and spatial scales of current forecasts are such that high-resolution forecasts are not practically possible” The need for integration of seasonal forecasts into process is identified as a key area. A lack of understanding of seasonal forecasts is identified as another reason for limited uptake by industry.

Forecasting Water Supply for the City of Cape Town

In the context of water management in the CoCT the application of regional seasonal forecasts for water supply management has not been utilised and instead rainfall from 2017 was used as a proxy for future dam level prediction (*Summary – Water Outlook*, 2018; pg. 4) thus providing the CoCT with a very conservative model due to the low rainfall experienced in 2017. Ziervogel et al., 2010, (Pg. 550) expressed that current seasonal forecast information is not meeting the needs of water resource managers and as such different formats of seasonal forecasts need to be explored to make seasonal forecasts more accessible to decision makers. Key findings from the study included that there is little evidence of seasonal forecast being used in the water management sector and that utilisation of seasonal forecast is hindered by a lack of understanding. Tailored products that are better integrated into water resource planning is suggested as a way to improve forecast utilisation.

Wolski et al., 2017, (Pg. 32) Suggest that developing hydrological seasonal forecasts would have the potential to both increase the skill of seasonal forecasts, due to the greater dependence on initial state of hydrological systems and would serve the dual purpose of providing water managers with a tailored product for use in decision making. Yet despite promising results from initial research no formal or practical activities have taken place. Thus Wolski et al. encourage co-exploratory activities in climate and hydrological prediction that would elude to research questions and improve understanding and ability to produce such forecasts. With this in mind the factors influencing the usefulness of a regional tercile based precipitation forecast for prediction of water storage in the Western Cape are here considered. These factors include identification of spatial variations of rainfall and differential rainfall regime influences. Further the relationship between streamflow and rainfall is investigated with the view to identifying the primary sources of uncertainty when a full blown hydrological model is linearly approximated.

Data Availability

A wealth of rainfall observations exists across the western cape. A total of 151 stations from SAWS and DWS were initially considered for this study. To ensure little bias existed within the results, only stations with more than 90% validity were considered after the data was processed and cleaned, and outlying variables flagged and removed through CSAG cleaning. Missing values were then interpolated from neighbouring points using a linear interpolation technique on a station by station basis. Further, only stations which spanned the 50-year period of 1967 to 2016 were considered. Following this process, a total of 27 stations were considered. It is worth noting that of the 99 DWS stations considered, only 16 were more than 90% valid, and of these none spanned the required 50-year period. The SAWS station data performed better with 27 of the 52 stations considered being both more than 90% valid and spanning the required 50-year period.

Collation and checking of data on water supply data has proved to be a complex process given the intricate nature of the WCWSS. Whilst the CoCT is by far the biggest user dependant on the system it is not the only user; the greater communities of Paarl, Wellington and Stellenbosch, urban users and irrigators in the Riviersonderend catchment region, as well as irrigators along the Berg and Eerste river systems are also dependent on supply from the WCWSS. This is further complicated as whilst the DWS is responsible for overlooking the management of the system as a whole, the Berg river dam is operated by DWS on behalf of the Caledon Tunnel Authority who own the dam, and the Wemmershoek, Upper Steenbras and Lower Steenbras dams are owned and operated by the CoCT. ("Reconciliation Strategy for the Western Cape Water Supply System", N.D.)

A potential solution to this is to consider the system as a whole, rather than in parts, allowing for a 'black box' approach to quantifying net water accumulation in the system. Whilst raw data on net water supply from the system does not appear to be available, the Water Outlook 2018 (Summary – Water Outlook, 2018) report, as well as the Current Water Supply Situation Business Briefing (Flower, 2017) from the CoCT include daily water production values. This therefore provides a method for quantifying net urban water use across the WCWSS. Values for agricultural use proved harder to quantify given the various stakeholders involved and the highly seasonal nature. However, data regarding agricultural use was available from Aurecon, the managers of the WCWSS water management. This, along with the CoCT reports on water usage were adapted for use in the CSAG big six dam monitor and have been made available for the purpose of this study. Thus, by introducing an evaporation parameter, dam inflows could be quantified.

The Water Resources of South Africa, 2012 Study (WR2012) provides a platform for the collation of data regarding the state of water supply in South Africa, Lesotho and Swaziland. Here, among other

data, audited dam records and balances are available. These records include monthly data of: gauge reading, contents, total outflow, irrigation outflow, industry and town supply, gross evaporation losses, inflow due to rain, calculated streamflow and unaccounted losses, thus providing a comprehensive resource for 150 dams across the South African region. However, these records only cover three of the six major dams that supply water to Cape Town, where the Berg river dam, Upper Steenbras dam and Wemmershoek dam are not included in this data. Thus, for the purposes of this study, the 19 years of comprehensive reservoir record that is/was available for Theewaterskloof dam, the largest dam in the WCWSS, was used as a proxy for the WCWSS as a whole. Within this dataset a rainfall value that quantifies net inflow from direct rain over the dam is provided. Here all stations available within the catchment are considered, thus this value is seen to be more robust than the SAWS Theewaterskloof station data and as such is used to quantify the rainfall-streamflow relationship.

Methodology

Spatial Analysis of Precipitation

To analyse to what extent rainfall variability exists within the western Cape and more specifically the catchment regions of the big six dam spatial analyses are undertaken. The data from an initial set of 151 precipitation stations from SAWS and DWS was processed and cleaned, and outlying variables were flagged and removed. Stations which did not span the period 1967-2016 were then excluded from the data set and in line with that done in a similar study by du Plessis & Schloms (2017) stations with less than 90% validity were excluded.

Time series plots of average daily precipitation across the stations and cumulative monthly and yearly rainfall plots were created to identify any trends or outlying points within the data set and to better analyse the nature of rainfall in the region. Here missing values were excluded from the analysis, however to avoid excessive complications in subsequent analysis missing values were set to zero.

Initial cluster analyses using a variety of clustering techniques including single linkage, average linkage and Ward's method linkage were undertaken. Ward's method proved to maximize the between group linkages when compared to the other methods and is the most frequently used hierarchical clustering technique for climatic classification. Where at each clustering step, the within-cluster variance is minimised and the similarity between the two-clusters is the sum of the variance between the two-variable summed over all the variables (Kalkstein, Tan & Skindlov, 1987). There is no clear trend in the data and standardization had little effect, as such standardized data is not considered.

A principle component analysis was then undertaken on the data. In order to further group stations into different rainfall zones as well as to determine whether any underlying pattern existed within the data. The initial results of the output proved to successfully differentiate the data and as such no rotations were considered. The results of principle component 1 and principle component 2 were ranked evenly about the range of the factors, with outlying factors placed into separate groups and remaining factors placed into 3 groups of even ranges. These groups were then colour coded and mapped to better visualize the distribution of the data.

In order to further compare results and verify the accuracy of the CRU3.22 reanalysis data a spatial PCA was conducted on this spatial precipitation data and compared to that of the station data. The distributions were deemed similar and thus to further analyse the distribution of rainfall over the

region the spatial precipitation data was plotted as six distinct nodes using a self-organizing map technique.

Rainfall – Streamflow Relationship

To consider to what extent rainfall events result in streamflow events and whether the same level of seasonality is seen in streamflow as is seen in rainfall the relationship is further explored. Using reservoir record data from the WR2012 assessment, monthly data on streamflow and rainfall were considered over the Theewaterskloof dam. Data from before 1991/11 was not considered as these values appeared to be inconsistent with the later data, where months before 1991/11 have inflow values in the order of 4 times smaller. Months with missing rainfall or streamflow data were excluded from the data set and thus a total 218 months of the possible 228 months are considered. Average monthly values of streamflow and rainfall were plotted to visualise the data.

After assessing the correlation between rainfall and streamflow to be sufficiently high at 0.719, density plots were considered to study the distributions of rainfall and streamflow and were deemed approximately normal and as such an initial single linear regression model was constructed over the 19-year period and modelled values compared to actual values. The inability of this method to capture the extent of seasonality within the data was noted and as such distinct linear models were constructed for each month and the changing relationship analysed by comparing the relative model's correlations, intercepts, coefficients, R-squared values and standard errors. The performance of this monthly model was again assessed through comparing modelled against actual values and through scatter plots.

To further assess the relationship between rainfall and streamflow in the case of tercile based seasonal forecasting the data was separated into seasons of Spring (SON), summer (DJF), autumn (MAM) and winter (JJA). The rainfall and streamflow values were then scaled as a fraction of their maximum values and plotted as an Empirical Cumulative Distribution Function where the percentiles are represented on the y-axis and scaled values on the x-axis, thus allowing comparison of the relative distributions.

Results and Discussion

Western Cape Rainfall Overview

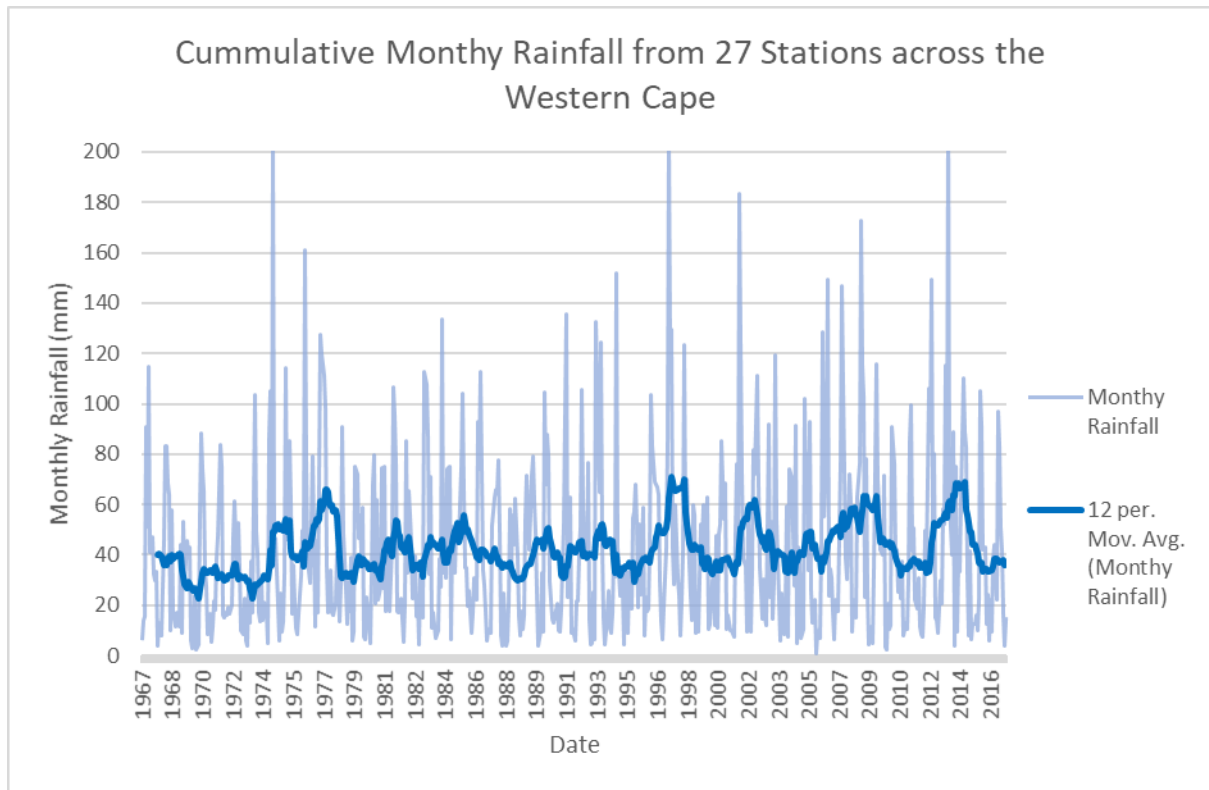


Figure 1: Monthly Cumulative Rainfall of mean rainfall from 27 stations across the Western Cape. Where a 6- month moving average of the monthly data is plotted with monthly cumulative data plotted in light blue.

The CoCT and surrounding water source areas are located in the south western cape rainfall region and thus receive predominantly winter rainfall (Engelbrecht et al., 2015). Figure 1 shows the extent of rainfall seasonality with peak rainfall values in the order of 100-200 mm/month occurring in Austral Winter (JJA) and minimum values of 2-10 mm/month occurring in Austral Summer (DJF) as observed from 27-stations across the Western Cape. No clear trend is observed in the data, where when yearly cumulative totals of the daily mean rainfall across the stations is considered an increase of 2.41mm/year over the 50-year period is observed, however if 2017 and 2018-up-to-present data is considered, which like 2016 were anomalously dry this trend will likely change. Thus, there is no long-term trend in rainfall in the Western Cape, however clear wet or dry spells of 2-4 years can be seen in Figure 1. Rainfall in the Western Cape is also highly event driven with rain only occurring on 56% of days across the 27 stations.

Spatial Analysis of Precipitation

Cape Town receives its water supply from six major dams, commonly referred to as the 'Big Six', these dams being: Voëlvlei dam, Wemmershoek dam, the Berg River dam, Theewaterskloof dam,

and the Upper and Lower Steenbras dams. These dams are located in the cape fold belt mountains to the North and East of the City, to supply the gravity fed water supply system and with the benefit of these regions being identified as watersheds (WWF, 2013) as they receive more rainfall than the surrounding coastal plains and Karoo hinterland regions. The locations of these dams are shown in Figure 2.



Figure 2: Map showing the location of the 'Big Six' dams (Green Marker) and the 27 precipitation stations considered (White Circle).

The extent of spatial rainfall variability across this rainfall region is initially analysed through a clustering technique of the 27 stations across the South Western Cape.

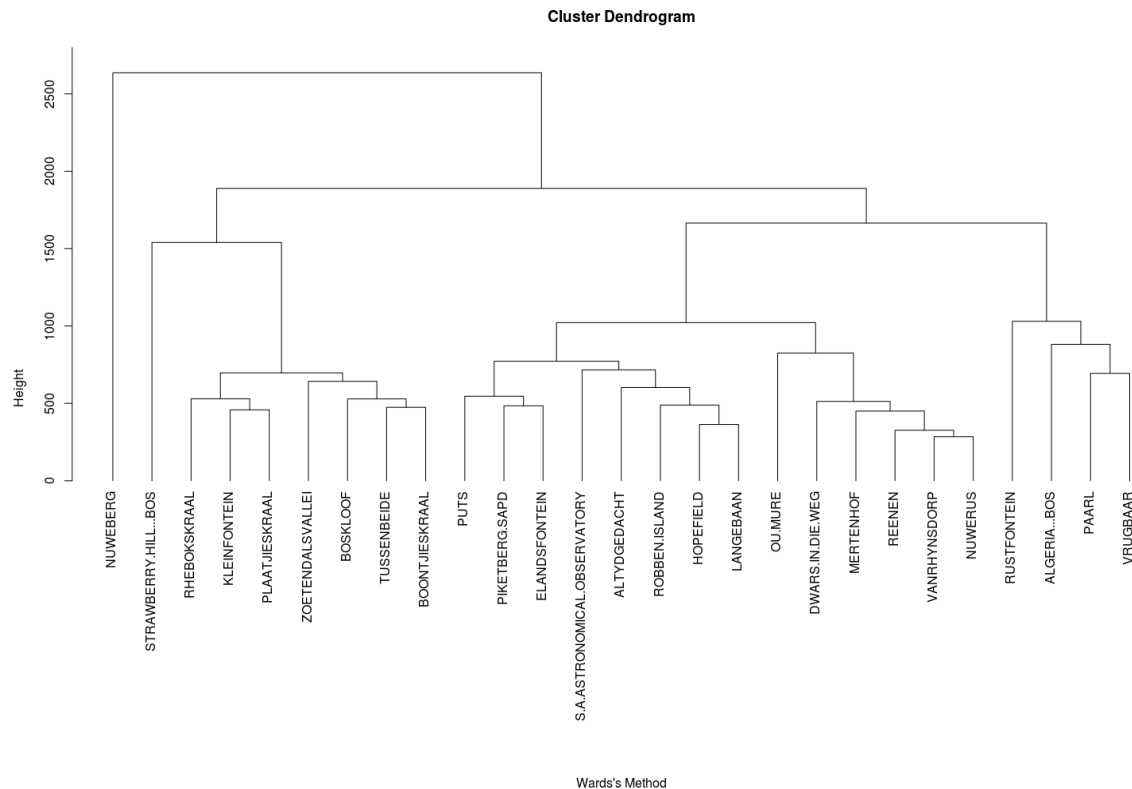


Figure 3: Cluster Dendrogram showing Euclidean linkage distance between stations after a Ward's method clustering technique is applied.

Figure 2, shows the linkage distance of clusters using the Ward's minimum variance clustering method. Whilst clear groupings are seen it is noted that the minimum linkage distance within clusters of approximately 500 is relatively large compared to the maximum linkage distance of 2500. This is potentially as a result of the high resolution of the data and the timing of rainfall events not being synchronous across the region. The groupings are however seen to be closely linked to the locations of the stations, with the Langebaan station closely linked to the Hopefield station where both stations are in the west and relatively far from any mountain ranges, as well as being only 28km apart. Similarly, a maximum linkage distance exists between the Hopefield station and the Newuberg station which are both on a similar longitude, however the Nuweburg station is located a further 130 km to the South and in a mountainous region. Close linkage distances are observed between S.A. Astronomical Observatory and Altydgedacht which is expected as the stations are all in close proximity to each other within the CoCT, the inclusion of the Robben Island, Hopefield and Langebaan stations does however appear to be anomalous due to these station being located some 100 km's apart, however this could be as a result of these stations being on a similar longitude and all being located on the western coast. Elandsfontein, Piketberg and Puts are also linked to these stations with only a small increase in linkage distance, this being as these stations are again located

along the west coast but this time even further North. Another grouping of similarly small linkage distances of <800 includes: Rhebooskraal, Kleinfontein, Plaatjieskraal, Zoetendalsvallei, Boskloof, Tussenbeide and Boontjieskraal. Again, these stations are all in relatively close proximity and this time all situated in the southern cape region, suggesting a distinct rainfall regime exists here. Stations where large linkage distances of >800 exist between them and any other stations and where no defined groupings exist include Rustfontein, Algeria – Bos, Strawberry Hill -Bos and Nuweberg. These are all seen to be stations located in mountainous regions thus suggesting rainfall in these regions is more variable and less dependent on the proximity of stations to one another than in coastal plain and Karoo regions, with localised feature having more influence on rainfall in these mountainous regions.

To further investigate to what extent different rainfall regimes can be identified within the Western Cape a principle component analysis is undertaken.

Table 1: Factor loadings and explained variance of principle components of daily precipitation data from stations across the western cape.

Station Name	PC1	PC2	PC3	PC4	PC5
NUWEBERG	12.26031932	2.367340658	5.548144878	-0.580784466	0.342201908
RUSTFONTEIN	6.334982915	0.45236041	0.044331586	1.348192322	0.459066798
ALGERIA...BOS	5.705319288	0.986646352	-2.783148449	-1.630954072	2.193292902
VRUGBAAR	5.267055545	0.725600614	-1.851207479	-0.24589779	-1.323168025
PAARL	4.356818275	1.100462492	-1.80491944	-0.64169209	-2.859584611
ELANDSFONTEIN	3.98297454	0.818350785	-1.983306855	-1.115240123	0.84752513
ALTYDGEDACHT	3.516768927	0.433368713	-0.897669164	0.302835368	-1.541473887
TUSSENBEIDE	3.397514212	-1.84191127	-0.499492558	2.112104754	0.042251395
PIKETBERG.SAPD	3.215758805	0.360726071	-1.184929935	-0.399710505	0.314076235
BOSKLOOF	3.058980477	-1.968229957	-0.254258211	1.592299406	0.241901999
STRAWBERRY.HILL...BOS	2.932999454	-7.426927223	0.947935501	-3.35110435	-0.718154987
S.A.ASTRONOMICAL.OBSERVATOF	2.860276601	0.32491166	-0.93104213	0.431582801	-2.142578573
BOONTJIESKRAAL	2.730511563	-1.32097686	-0.445277975	1.575237348	0.100377949
ROBBEN.ISLAND	2.609142726	0.338495522	-0.625723594	0.253433064	-1.020536516
PUTS	2.574199878	0.513282218	-1.456302454	-0.756956159	0.994913823
ZOETENDALSVALLEI	2.373411908	-1.739421283	-0.193613806	1.556427457	0.220451954
RHEBOKSKRAAL	2.351290142	-1.865328155	-0.357821991	1.135201641	0.464202187
HOPEFIELD	2.261820324	0.296928107	-0.839150193	-0.071776738	-0.007888197
LANGEBAAN	2.103849074	0.27374878	-0.753611914	0.017318934	-0.285531928
PLAATJIESKRAAL	2.082904011	-2.630311929	-0.222551137	1.358491502	0.245318969
MERTENHOF	2.010597671	0.410457179	-1.070132293	-0.586655511	0.867802736
KLEINFONTEIN	1.975331796	-2.313681256	-0.233565852	1.116805068	0.361647448
OU.MURE	1.481776221	-0.448695191	-0.765523683	0.323337471	1.599163419
VANRHYNSDORP	1.116326244	0.102199136	-0.670700584	-0.246357235	0.447449225
NUWERUS	1.082950383	0.011165668	-0.572550343	-0.004346233	0.437601597
DWARS.IN.DIE.WEG	0.894762407	-0.631682077	-0.27907544	0.026203313	0.370633799
REENEN	0.849011327	0.059317194	-0.441335707	-0.170067934	0.452192288
Explained Variance (%)	49.8	11.52	7.42	4.37	3.69

Principle component 1 explains 49% of the total variance within the data, significantly more than is seen from Factor 2, which explains 10% of the total variance. The first component is unimodal and similarly to that of the cluster analysis, the LangebanWeg station and the Hopefield station are linked closely with very similar factors loadings, whilst Dwars-in-die-Weg, a station located to the East of the fold-belt mountains, and the foothill station of Vrugbaar have a large difference between factor loadings in PC1.

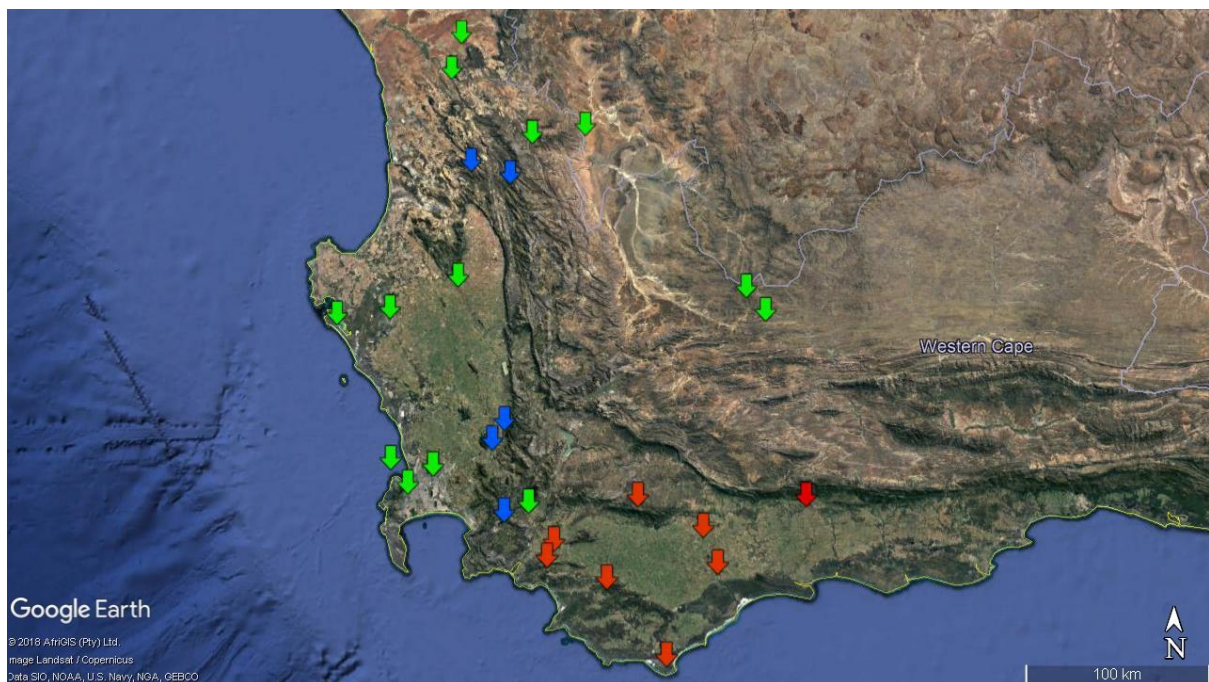
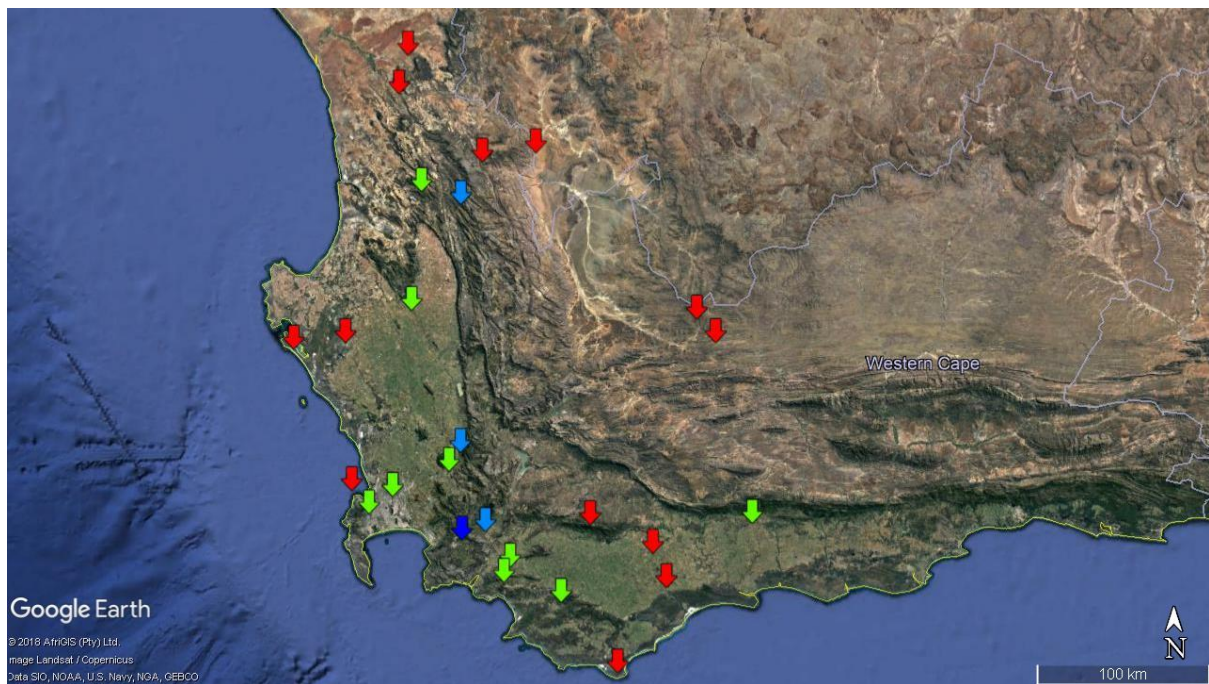


Figure 4: Factor loadings from PC1 (above) and PC2 (below) plotted in distinct groups as a function of their factor loadings where blue corresponds to high loadings, green to intermediate and red to low factor loadings.

Figure 5, shows the stations grouped into 3 distinct groups as a function of their factor loadings, with outlying stations placed into separate groups to not mute any signal from the factor loadings. Where blue corresponds to high factor loadings, green to intermediate and red corresponds to low factor loadings. The groups produced by principle component 1 appear to be highly influenced by their location relative to the Cape Fold belt mountains with high factor loadings occurring in stations located on the western slopes of the fold belt mountains, intermediate loadings occurring in stations on the western foothills of the fold belt mountains and low factor loadings occurring in coastal stations and stations to the east of the fold belt mountains. This pattern is seen to be strongest in Northern stations, stations in the South Coast do not follow this trend as closely. Some influence from Table Mountain can be seen in stations immediately to the west of the mountain. Thus principle component 1 seems to explain a function of the total quantity of precipitation received at stations during rain events where due to the Foehn effect regions located at the foothills and within mountainous regions, synchronous with the location of the 'big six' dams, receive more rainfall than regions located after mountainous regions (du Plessis & Schloms, 2017).

The spatial variation of the groups produced by principle component 2 seem to separate the stations along climatic regions, with stations with low factor loadings located in mountainous regions and stations with intermediate factor loadings in coastal and Karoo regions. Stations with low factor loadings clearly correspond to the south coast rainfall region, similarly identified by du Plessis & Schloms (2017). Thus, whilst the Western Cape is predominantly a winter rainfall region the borders between this and neighbouring regions is unclear and thus process from surrounding regions may influence rainfall to different extents across the region. The inclusion of distinct group with lower factor loadings in the South coast region is as a result of these south coast regions bordering the all-year round rainfall region and as such receiving comparatively more rain than the surrounding regions in summer (Engelbrecht et al., 2017). This could suggest dams located further to the south such as the upper and lower Steenbras dams are affected to a greater extent by the Southern Cape all-year round rainfall regime than the more Northern Voelvlei dam and thus may be dependant to a lesser extent on winter rainfall. Thus, a need for considering the region at an appropriate scale to fully capture these variations between the rainfall at various dams is identified.

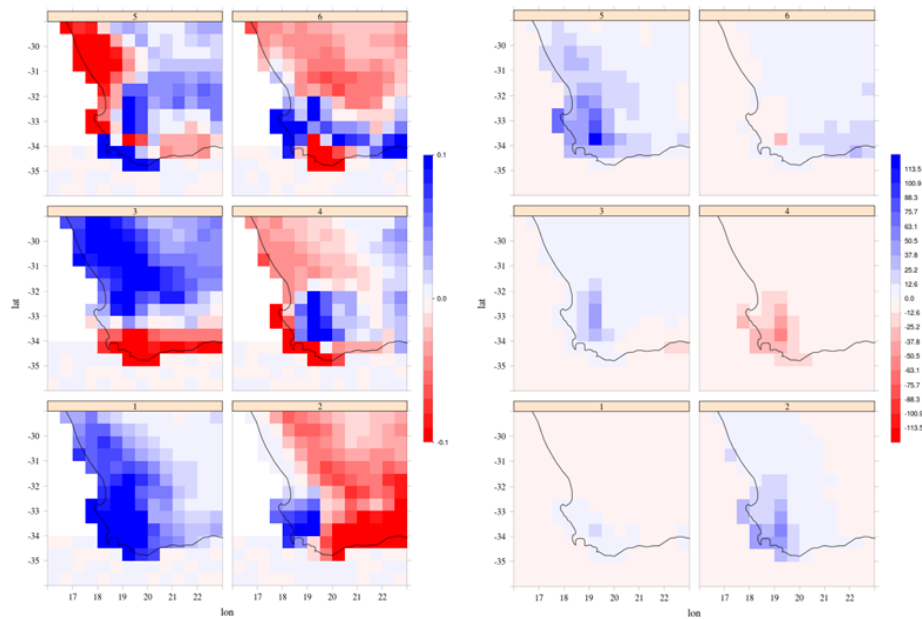


Figure 5: Factor loading of monthly precipitation data from the cru3.22 re-analysis model (left) and nodes of spatial rainfall patterns generated through self-organising map techniques of the same monthly precipitation data (right).

A principle component analysis was subsequently conducted on monthly precipitation data from the cru3.22 reanalysis model, shown in figure 6, to see if similar results are found, the relatively coarse resolution of this data at 0.5° is acknowledge and thus only used in conjunction with the results from the station data. Principle component 1 shows high factor loading in western regions and low factors in northern and eastern regions and is thus explaining the contrasting seasonal precipitation experienced in these regions, with western regions experiencing predominantly winter rainfall and northern regions receiving predominantly summer rainfall (Rouault & Richard, 2005) Principle component 2 explains the contrast between precipitation in the south coast and that of the surrounding regions, where the south coast regions receives comparatively more summer rainfall than surrounding regions. Similar results to that of the station data. The remaining factors show variations on smaller scales but are not considered due to the coarse resolution of the data. Nodes 1,2,3 and 5 in the self-organizing map shown in figure 6, identify increased precipitation in mountainous regions relative to surrounding regions (du Plessis & Schloms, 2017). Whilst node 6 identifies increases precipitation in the south coast regions, explaining a summer rainfall regime (Engelbrecht et al., 2015). Node 4 identifies drier conditions in the south western cape compared to that of surrounding regions and is thus potentially identifying another summer rainfall regime. Thus, here potentially more variations are seen within the mountainous region than surrounding regions although anomalies do appear to behave in a synchronous manner across the region.

Rainfall Streamflow Relationship

Although rainfall is the only driver of water input into Cape Town's dams, direct rainfall accounts for only 7% of inflow into the Theewaterskloof dam, the rest being from streamflow. Whilst it may seem obvious that a strong and linear relationship should exist between rainfall and streamflow into a dam as rainfall is the only water input into the system - as negligible snow or ice melt is experienced in the Western Cape - the reality is far more

complicated. Here this relationship is explored, where due to data

availability constraints the Theewaterskloof dam is used as a proxy for net water storage accumulation in the Western Cape. Figure 7 shows the monthly variation in rainfall and streamflow. Whilst both are clearly seen to be highly seasonal with peak values occurring in winter the seasonality seen in Streamflow is seen to be far more pronounced and is seen to lag peak rainfall with relatively lower values in spring; a largely synchronous peak occurring in July to August and then higher relative streamflow values seen in Autumn. This being as a result of changing soil moisture and evaporation rates serving to both increase the seasonality of streamflow and result in increased post winter and decreased pre-winter streamflow's. (Davie, 2008)

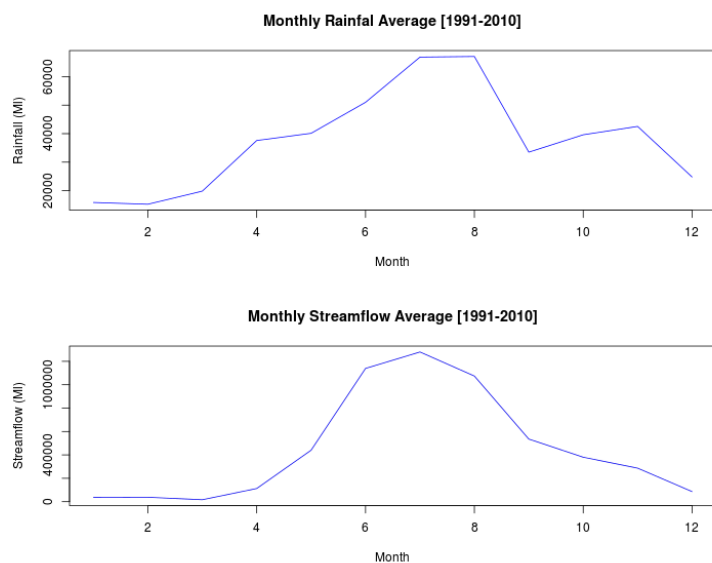


Figure 7: Average monthly values for streamflow and rainfall over the 19 years period 1991-2010

With the view to best capture the change in relationship between rainfall and streamflow as the seasons progress, distinct linear regression models are constructed for each month to quantify this changing relationship. The extent of variability capture by this model is shown as actual vs predicted values in figure 9, where through this technique a 90.88% correlation between modelled and actual values exists.

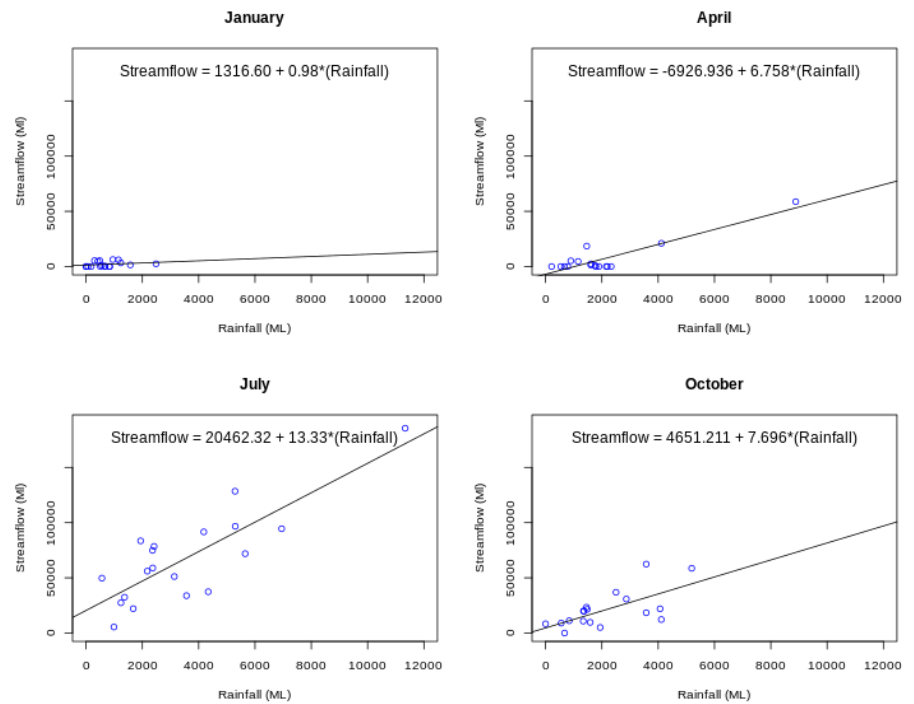


Figure 8: Scatter plots of selected monthly streamflow and rainfall with linear regression model equations plotted.

Table 2: Monthly linear regression model statistics

Month	Correlation	Regression Intercept (ML)	Coefficient of Rainfall	R-squared	Standard error (ML)
January	0.247	1316	0.98	0.061	900
February	-0.190	2881	-1.13	0.036	1346
March	0.145	528	0.3622	0.021	862
April	0.902	-6926	6.75	0.814	2320
May	0.582	6572	8.46	0.339	7367
June	0.865	10083	20.38	0.749	9473
July	0.823	20462	13.33	0.678	9647
August	0.474	35377	5.99	0.225	10346
September	0.690	7540	13.99	0.476	7029
October	0.657	4651	7.69	0.431	5650
November	0.605	9160	3.59	0.367	3372
December	0.297	3275	1.13	0.088	1610

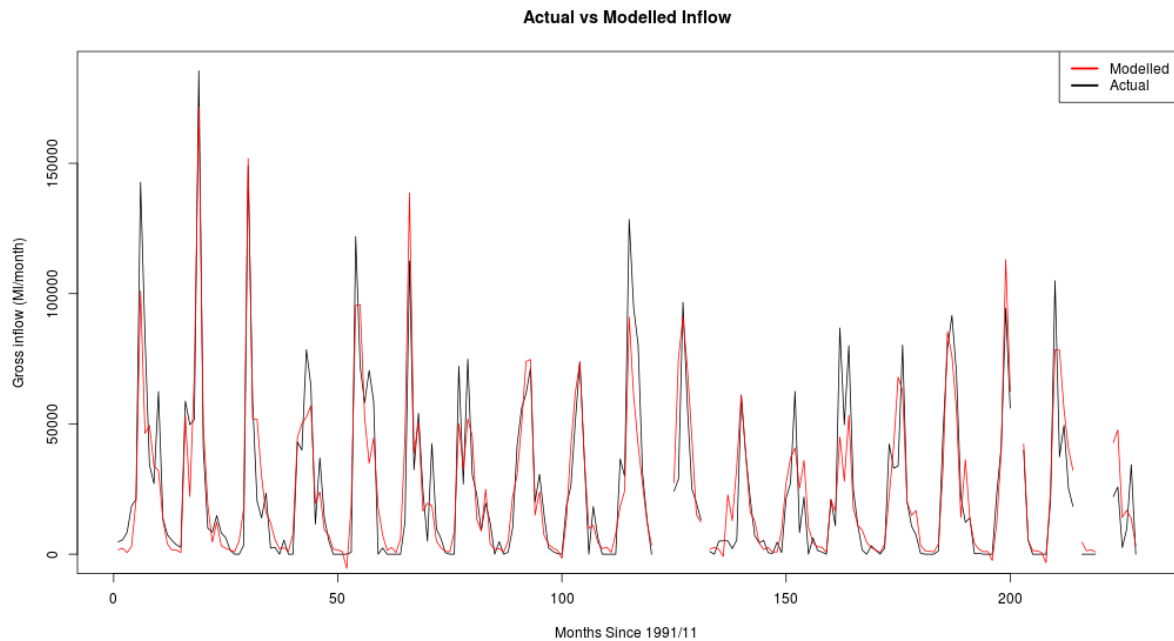


Figure 6: Actual Streamflow values into the Theewaterskloof dam plotted against those modelled using distinct monthly linear models.

Table 2 shows clear distinction across seasons, with increased correlations occurring in winter and autumn compared to that of summer and spring. Increased coefficients are also seen over these periods indicating rainfall in winter and autumn results in a proportionally larger amount of streamflow than events occurring in summer or spring. An anomalous relationship is seen in

February where a negative relationship exists between rainfall and streamflow, this is however attributed to error within the data as only 17 data points are available for this month and further due to very low streamflow values experienced producing an anomalous signal. Higher correlations and R-squared values are seen in winter suggesting a potentially more linear relationship existing in

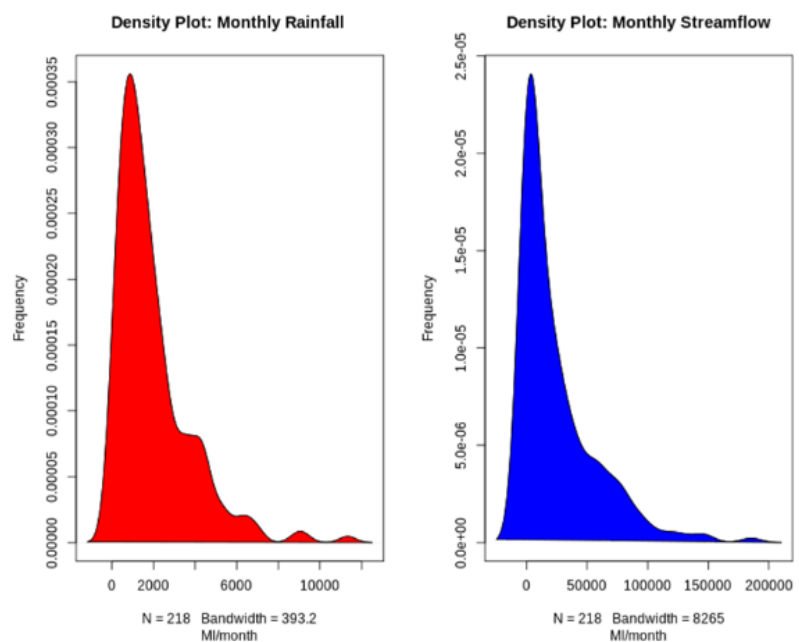


Figure 7: Density plots showing the distribution of rainfall and streamflow.

winter, however increased standard errors are also seen. Thus, whilst rainfall explains a larger percentage of the variance in winter, it is still not sufficient to only consider rainfall to model streamflow as these increased streamflow's bring about larger standard errors. Figure 8 further explains this behaviour where near zero streamflow values in January mean that little error can be associated with a linear approximation despite rainfall seeming to have little influence on these low streamflow values. The opposite is true in July where a clear positive and linear relationship can be seen between rainfall and streamflow, where peak rainfall values result in peak streamflow values. Thus, to make the seemingly straight forward assumption that increased rainfall results in increased streamflow's is not necessarily accurate. Where the season being considered has a large influence on these results such that in summer increased rainfall does not necessarily result in increased streamflow's whilst in winter this relationship does exist but is associated with large standard errors, however whilst larger standard errors are observed in July, the standard error of July is 5.35% of the observed range, whilst in summer the standard error is 14.2% of the observed range thus indicating that using rainfall as the sole predictor of streamflow is in fact more valid in winter than in summer.

The Western Cape relies primarily on cold fronts (Engelbrecht et al., 2015) for rainfall and as such results in a distribution that whilst is expectedly highly skewed toward low values, as is the case with streamflow, the rainfall distribution has an extended tail toward extreme events. This same behaviour is not seen in streamflow, suggesting these extreme rainfall events have a lesser influence on streamflow than other rain events. These distributions are shown in Figure 10. The implication of this being that the assumption that a below-normal rainfall (Below the 33% percentile) season may not correlate to a similarly below-normal streamflow season, being that streamflow is skewed to a lesser extent.

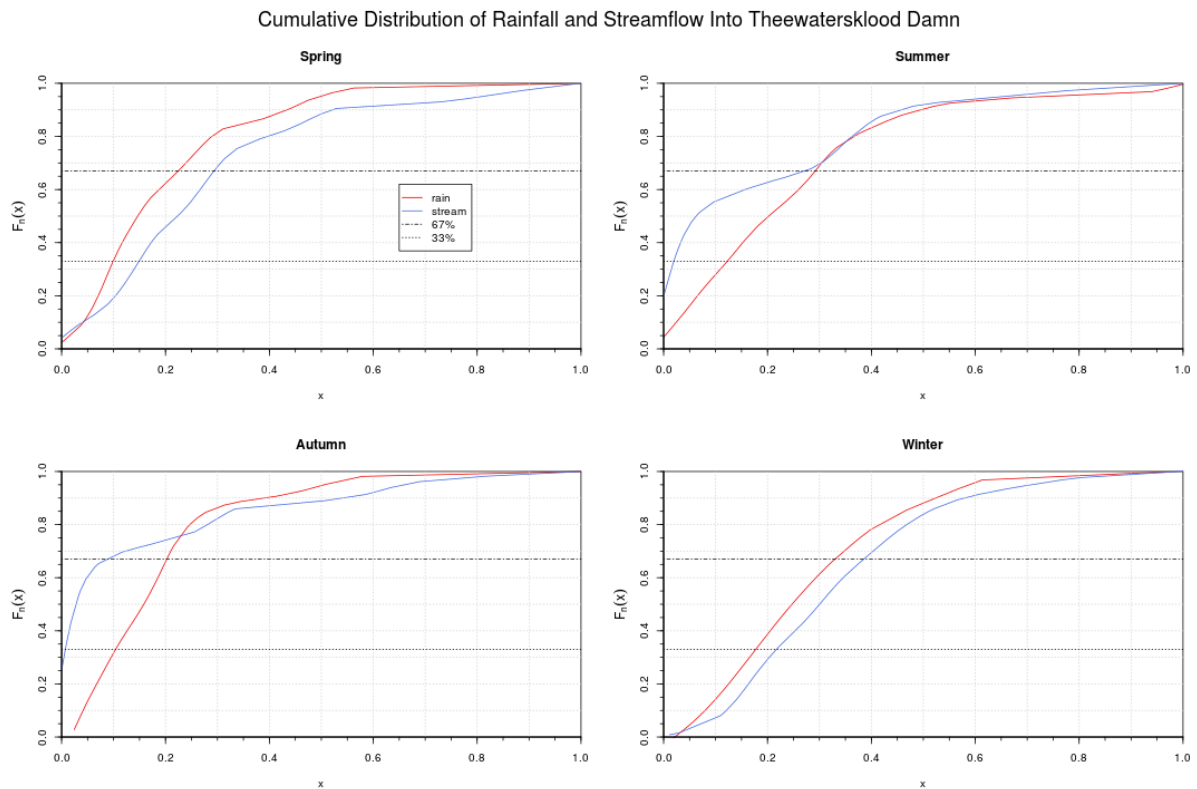


Figure 8: Empirical Cumulative Distribution Functions of scaled streamflow and rainfall values with the 1st and 2nd tercile divisions indicated.

This is further explored in figure 11, where scaled streamflow and rainfall values are considered against the divisions of terciles. The largest deviation is seen in Autumn (SON) where the 33% division of rainfall maps to a value of 0.1 of the maximum value, whereas 0.1 of the maximum value of streamflow maps to the 70th percentile. Indicating that a below-normal rainfall season only correlates to a streamflow season that is not above-normal. A similar relationship but to a lesser extent is seen in spring. In winter and summer however, an opposite relationship is seen where the 1st tercile of rainfall correlates to a lower percentile of streamflow, indicating that a below normal rain event relates to an even more severe low streamflow event. It is noteworthy that these disparate distributions of rainfall and streamflow appear to be stronger in shoulder seasons, where stronger soil moisture and land use fluxes occur. Thus, this often-made assumption that a below-normal rainfall season will invariably result in a below-normal streamflow season is an invalid assumption.

Conclusion

Rainfall in the Western Cape is seen to be highly seasonal with winter rainfall accounting for majority of the yearly total rainfall. The location of Cape Town's major dams within the Cape Fold Belt mountains is shown to be crucial because of the increased quantity of precipitation received within these watershed regions, where a classical Foehn effect is seen to occur during winter storm events. Further precipitation in Mountainous regions is seen to be more variable than that of Karoo and coastal plain regions potentially because of localised features and their interactions with storm tracks having a high influence on quantity of rain received.

Whilst the water catchment regions are all located within the Cape Fold Belt Mountains and are all within relatively close proximity to one another within the South Western Cape Winter Rainfall region, neighbouring rainfall regimes are seen to have variable influences across the region, where more southern dams, such as the upper and lower Steenbras as well as Theewaterskloof, are seen to be influenced to a greater extent by the South Coast all year rainfall region. Thus, to consider the dams as a single entity rather than as individual components of the system could introduce more uncertainty, further this spatial variation is significant such that a higher resolution seasonal forecast may better capture the variability within the catchment region than those traditionally considered.

Further the relationship between rainfall and streamflow is shown to be far more complicated than initially perceived, where a single linear approximation is not sufficient to fully describe the relationship. Here the correlation between rainfall is seen to be at a maximum during winter months, where high correlations are seen but still associated with large standard errors. In summer the relationship is far more complicated with very low correlations and relatively higher standard errors. The notion of a below normal rainfall month resulting in below normal streamflow month is shown to be invalid and highly dependent on the season considered. Where in shoulder seasons the 1st tercile of rainfall correlates to a far higher percentile of streamflow due to strong land use and soil moisture fluxes occurring in these seasons. Whilst in summer and winter the opposite, albeit to a lesser extent, is seen where the 1st tercile of rainfall correlates a lower percentile of streamflow. Such is this distributional shift that a below-normal rainfall forecast for autumn correlates to a streamflow season that is not above-normal. A similar relationship but to a lesser extent is seen in spring. The opposite but to a far lesser degree is seen in summer and winter where the opportunity exists for increasing the appreciable skill of a forecast by forecasting for a higher percentile of rainfall that will correlate to a below-normal streamflow event.

Thus, sources of added uncertainty when using a tercile based regional precipitation forecast for prediction of water storage are identified. Whilst some sources, such as the changing rainfall

streamflow relationship, distributional shifts and the need for considering rainfall at an appropriate resolution, present opportunities for increasing predictive skills, others are more a function of the nature of integrating a seasonal forecast into process and cannot necessarily improve the forecasts other than through their identification as sources of uncertainty.

Study Limitations

Whilst the spatial rainfall analysis was successful in differentiating rainfall along acknowledged rainfall regions in part due to the large size of the dataset considered, extended periods of missing data existed within the dataset, and the simplification of setting these values to zero may have had a significant impact. Thus, these results could have been improved through interpolating these missing points from reanalysis data or through excluding missing data from the spatial analysis entirely. An initial aim of the study was to undertake a skill analysis on a seasonal forecast for the prediction of dam levels. Where through, the regression model developed, a seasonal forecast would be used to differentiate between periods of above-normal, normal or below-normal streamflow and assessed against observed streamflow events to determine the skill of the forecast.

References

Davie, T. 2008. *Fundamentals of hydrology*. Routledge.

Descriptions of the IRI Climate Forecast Verification Scores. 2013. Available: <https://iri.columbia.edu/wp-content/uploads/2013/07/scoredescriptions.pdf>.

Engelbrecht, C.J., Landman, W.A., Engelbrecht, F.A. & Malherbe, J. 2015. A synoptic decomposition of rainfall over the Cape south coast of South Africa. *Climate Dynamics*. 44(9):2589–2607. DOI: 10.1007/s00382-014-2230-5.

Engelbrecht, C.J., Landman, W.A., Graham, R. & McLean, P. 2017. Seasonal predictive skill of intraseasonal synoptic type variability over the Cape south coast of South Africa by making use of the Met Office Global Seasonal Forecast System 5. *International Journal of Climatology*. 37(4):1998–2012. DOI: 10.1002/joc.4830.

Hewitson, B.C. & Crane, R.G. 2006. Consensus between GCM climate change projections with empirical downscaling: precipitation downscaling over South Africa. *International Journal of Climatology*. 26(10):1315–1337. DOI: 10.1002/joc.1314.

Johnston, P.A., Archer, E.R.M., Vogel, C.H., Bezuidenhout, C.N., Tennant, W.J. & Kuschke, R. 2004. Review of seasonal forecasting in South Africa. *Climate Research*. 28(1):67–82. DOI: 10.3354/cr028067.

Kalkstein, L.S., Tan, G. & Skindlov, J.A. 1987. An Evaluation of Three Clustering Procedures for Use in Synoptic Climatological Classification. *Journal of Climate and Applied Meteorology*. 26(6):717–730. DOI: 10.1175/1520-0450(1987)026<0717:AEOTCP>2.0.CO;2.

Mason, S.J., Joubert, A.M., Cosijn, C. & Crimp, S.J. 1996. Review of seasonal forecasting techniques and their applicability to southern Africa. *WATER SA-PRETORIA*. 22:203–210.

Mukheibir, P. & Ziervogel, G. 2007. Developing a Municipal Adaptation Plan (MAP) for climate change: the city of Cape Town. *Environment & Urbanization*. 19(1):143–158. DOI: 10.1177/0956247807076912.

New, M., Hewitson, B., Stephenson, D.B., Tsiga, A., Kruger, A., Manhique, A., Gomez, B., Coelho, C.A.S., et al. 2006. Evidence of trends in daily climate extremes over southern and west Africa. *Journal of Geophysical Research: Atmospheres (1984–2012)*. 111. DOI: 10.1029/2005JD006289.

Otto, F., Wolski, P., Lehner, F., Tebaldi, C., van Oldenborgh, G.J., Hogesteger, S., Singh, R., Holden, P., et al. 2018. Available: <https://www.worldweatherattribution.org/the-role-of-climate-change-in-the->

2015-2017-drought-in-the-western-cape-of-south-africa/.

Philippon, N., Rouault, M., Richard, Y. & Favre, A. 2012. The influence of ENSO on winter rainfall in South Africa. *International Journal of Climatology*. 32(15):2333–2347. DOI: 10.1002/joc.3403.

du Plessis, J.A. & Schloms, B. 2017. An investigation into the evidence of seasonal rainfall pattern shifts in the Western Cape, South Africa Title. *Journal of the South African Institution of Civil Engineering*. Vol 59 No. Available: <http://dx.doi.org/10.17159/2309-8775/2017/v59n4a5>.

“Reconciliation Strategy for the Western Cape Water Supply System”. n.d. Available: http://www.dwa.gov.za/Projects/RS_WC_WSS/sa.aspx.

Rouault, M. & Richard, Y. 2005. Intensity and spatial extent of droughts in southern Africa. *Geophysical Research Letters*. 32(15). DOI: 10.1029/2005GL022436.

Simpkins, G. 2018. Running dry. *Nature Climate Change*. 8(5):369. DOI: 10.1038/s41558-018-0164-3.

staff", "GroundUp. 2018. Facts and myths about Cape Town’s water crisis. *GroundUp*. Available: <https://www.groundup.org.za/article/facts-and-myths-about-cape-towns-water-crisis/>.

Summary – *Water Outlook 2018*. 2018. Water Institute of South Africa. Available: <http://wisa.org.za/wp-content/uploads/2018/01/Water-Outlook-2018-Summary.pdf>.

Unknown. 2015. *WRC Drought Factsheet 1 - Background to current drought situation in South Africa*. Available: https://www.gov.za/sites/www.gov.za/files/gcis_documents/dourght-background-wrc.pdf.

Vogel, C.H. & Drummond, J.H. 1993. Dimensions of Drought: South African Case Studies. *GeoJournal*. 30(1):93–98. Available: <http://www.jstor.org/stable/41145720>.

Wolski, P. 2018. How severe is Cape Town’s “Day Zero” drought? *Significance*. 15(2):24–27. DOI: 10.1111/j.1740-9713.2018.01127.x.

Wolski, P., Lennard, C., Jack, C. & Tadross, M. n.d. Seasonal hydrological forecasting - current state of play. *September/October 2017*. 30–33. Available: http://journals.co.za/docserver/fulltext/waterb_v16_n5_a6.pdf?expires=1521109297&id=id&accname=57709&checksum=C5131A4F1CBA9086C30FCBE0DE8D6B92.

WWF. 2013. Available: http://dtnac4dflyw8.cloudfront.net/downloads/wwf_sa_watersource_area10_lo.pdf?9322/An-Introduction-to-South-Africas-Water-Source-Areas.

Ziervogel, G., Johnston, P., Matthew, M. & Mukheibir, P. 2010. Using climate information for supporting climate change adaptation in water resource management in South Africa. *Climatic Change*. 103(3):537–554. DOI: 10.1007/s10584-009-9771-3.

van Zyl, P. 2017. TRENDS IN WESTERN CAPE URBANISATION. In *Presentation to the Western Cape Property Development Forum Conference 2017*. Western Cape Government. Available: http://www.wcpdf.co.za/2017_conference/3 - PietvanZyl.pdf.